Parastoo Pilevar

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Research Profile

Ph.D. student in Electrical and Computer Engineering working on efficient and robust deep learning for vision–language models (VLMs) and vision–language–action models (VLAs) on edge and robotic platforms. Current research centers on model quantization, out-of-distribution robustness, and the accuracy, memory, and latency trade-offs that govern deployment-constrained perception.

Education

University of Maryland, College Park — 2025 – Present
Ph.D. in Electrical and Computer Engineering; M.S. expected 2026 | Advisor: Ang Li — *College Park, MD*

- GPA: 3.9/4.0. Coursework: Advanced Numerical Optimization, Selected Topics in ML, Convex Optimization, Information Theory, Estimation and Detection Theory, Advanced Digital Signal Processing.

Sharif University of Technology — 2015 – 2020
B.S. in Electrical Engineering; GPA: 3.61/4.0 | Advisor: Mohammadreza Pakravan — *Tehran, Iran*

- Thesis: hardware tester for JTAG chain, Ethernet, and E1 interfaces. Coursework: digital signal processing, analog circuits, communications systems, numerical methods.

Research Experience

Graduate Research Assistant — 2025 – Present
University of Maryland, College Park — CASE Lab — *College Park, MD*

- Designed and ran systematic post-training quantization (PTQ) experiments across multiple VLM architectures to study calibration-data sensitivity, showing that modality-skewed calibration sets cause measurable accuracy degradation in compressed models.
- Implemented a feature-space rectification method for quantized vision backbones on SWaP-constrained (size, weight, and power) edge platforms; trained a small low-rank adaptation (LoRA) adapter on the frozen 4-bit backbone using source-domain data only, recovering most of the PTQ robustness gap under sensor noise and weather corruption.
- Built PTQ evaluation pipelines tracking accuracy, memory footprint, and latency trade-offs across 4-bit and 8-bit configurations on standard robustness benchmarks (ImageNet-C, PACS-style shifts).

Research Assistant — 2022 – 2023
University of Tehran — Communication Networks Research Group | Supervisor: P. Shariatpanahi — *Tehran, Iran*

- Designed and evaluated cache-replacement policies for edge networks using federated reinforcement learning and federated multi-armed bandit frameworks.
- Investigated privacy-preserving learning strategies in distributed multi-agent systems.

Publications and Manuscripts

Recti-Q: Feature-Space Rectification for OOD-Robust Quantized Perception in Edge Robotics

- H. Yaghoubi*, **P. Pilevar***, M. Lin (*equal contribution). *IROS 2026.*
  Identifies and quantifies the out-of-distribution (OOD) robustness gap introduced by 4-bit PTQ in large vision backbones on resource-constrained robotic platforms, and proposes a frozen-backbone LoRA adapter trained on source data only that preserves PTQ memory savings while closing most of the robustness gap.

On the Sensitivity of Data-Driven Quantization in Vision–Language Models

- Z. Liu, S. He, **P. Pilevar**, A. Li. *Manuscript in preparation.*
  Systematic study of calibration-data effects in VLM PTQ across architectures and quantization methods, demonstrating that task- and modality-aligned calibration is a critical and underexplored axis of multimodal model compression.

Teaching Experience

Teaching Assistant — Signals and Systems — 2025 – Present
University of Maryland, College Park — *College Park, MD*

- Led discussion sections, wrote and graded problem sets, and held office hours for undergraduate engineering students.

Technical Skills

Programming Languages: Python, MATLAB, C, SQL, Bash
ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn
Data & Scientific Computing: NumPy, pandas, SciPy, PySpark, Matplotlib, Seaborn
Systems & Tools: Linux, Git